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NMAP to Discover live Host using Nmap Scans (ARP, ICMP, TCP/UDP) on the Try.Hack.Me platform.

* Aim:

To study the working of the Nmap tool and discover live hosts in a network using different host discovery techniques such as ARP Scanning, ICMP Scanning, TCP and UDP scanning on the Try Hack Me Platform. The experiment also aims to analyse the behaviour of different protocols and compare the results obtains from various scanning method.

* Step :-

1) Open the Try Hack me platform and select the Nmap room.

2) Study the network topology and identify the source and destination hosts.

3) Use ARP requests to discover the MAC Address for local Network.

4) Observe how ARP packets are broadcast with the network and identify the devices the receive the request.

5.) Use the ICMP protocol to send ping request between two System and observe the response.

6) Analyse the packet flow and identify the exchange of ARP packets before ICMP communication.

7) Perform ARP host discovery using the Nmap command and identify all active hosts in the local network.

8) Perform ICMP Echo scanning and observe the host that respond to the scan.

9) Perform ICMP Timestamp Scanning and analyse the response generated by target system.

10) Perform ICMP Address mask scanning and compare the result with the previous scans.

11) Perform TCP Syn Scanning to identify live hosts by analysis SYN and ACK packets.

12) Perform TCP ACK Scanning to verify the accessibility of the target hosts.

13) Perform IP ACK Scanning to verify how Nmap identifies active hosts through UDP Packets.

14) Perform a Reverse DNS lookups to obtain the hostname of the rediscovered system.

15) Compare the result obtained from ARP, ICMP, IP and UDP Scanning.

* Result:

Thus, live hosts in the network have successfully discovered using ARP, ICMP, TCP and UDP Scanning techniques with the Nmap tool.